🏋️ FITPULSE FITNESS TRACKER

Smart Fitness & Workout Tracking Application

A Full-Stack Web Platform to Log Workouts, Monitor Progress, and Set Fitness Goals

Tech Stack: PHP 8 • MySQL 8 • Bootstrap 5 • Chart.js

BACKGROUND

Problem Statement & Project Motivation

The Real-World Problems

• Workout Inconsistency: Most fitness attempts fail due to lack of habit tracking and accountability.

• Manual Logs are Inefficient: Paper diaries or notes apps are tedious, easily forgotten, and lack visual feedback.

• Bloated Commercial Apps: Existing solutions push paywalls, expensive subscriptions, and invasive ads.

• Data Disconnection: Workouts, weight history, and targets are rarely integrated in one clean dashboard.

Our Solution: FitPulse

• Centralized Web Platform: A lightweight, responsive web app accessible on any desktop or mobile browser.

• Automated Calculations: Automatically computes calories burned (via MET formula) and body metrics (BMI, BMR, TDEE).

• Interactive Visualizations: Real-time progress curves and calorie distribution charts.

• Goal Accountability: Dynamic milestone bars that auto-update as you exercise.

CORE REQUIREMENTS

Project Objectives & Scope

1. Log Workouts

- Record activities across 9+ sports & gym categories.
- Capture duration, intensity, distance, and custom notes.
- Log specific strength exercises with sets, reps, and weights lifted.
- Scientific MET calorie burn auto-estimation.

2. Monitor Progress

- Interactive 7-day activity & calorie expenditure charts.
- Category distribution doughnut graphs.
- Historical body weight tracking curve vs target weight.
- Monthly workout volume metrics and weigh-in log.

3. Set Fitness Goals

- Target weight loss & muscle gain milestones.
- Weekly workout frequency and calorie burn targets.
- Automated progress percentage tracking.
- Achievement celebration badges and statuses.

TECHNICAL DESIGN

System Architecture & Technology Stack

Frontend Layer

- HTML5, CSS3, JavaScript (ES6+)
- Bootstrap 5.3: Responsive layout, mobile-ready UI grid.
- Chart.js: Hardware-accelerated canvas charts (bar, line, doughnut).
- FontAwesome 6: Crisp visual icons.

Backend Layer

- PHP 8.3: Fast, modern server-side processing.
- PDO (PHP Data Objects): Secure parameterized queries protecting against SQL injection.
- Bcrypt Hashing: password_hash() for robust user credentials security.
- Session Engine: State management across pages.

Database Layer

- MySQL 8.0: Robust relational database management.
- InnoDB Storage Engine: ACID compliant transactions.
- Referential Integrity: Foreign keys with ON DELETE CASCADE constraints.
- Optimized Indexing for fast search and aggregation.

DATABASE ARCHITECTURE

Relational Database Schema Design

Core Entity Tables

• users: ID, Name, Email, Password Hash, Age, Gender, Height, Current Weight, Target Weight, Daily & Weekly Goals.

• workouts: ID, User ID (FK), Title, Category, Workout Date, Duration (mins), Calories Burned, Intensity, Distance (km), Notes.

• workout_exercises: ID, Workout ID (FK), Exercise Name, Sets, Reps, Weight Lifted (kg).

Progress & Tracking Tables

• weight_logs: ID, User ID (FK), Weight (kg), Log Date, Notes (maintains full historical body composition curve).

• fitness_goals: ID, User ID (FK), Title, Goal Type, Target Value, Current Value, Unit, Start & Target Dates, Status.

• Relational Design: Clean 1-to-Many relationships linking workouts, weights, and goals directly to each user account.

SCIENTIFIC LOGIC

Algorithms & Physiological Computations

MET Calorie Estimation Formula

- Formula: Calories = Duration (min) × [(MET × 3.5 × Weight in kg) / 200]
- Running MET: 9.8 | HIIT MET: 8.5 | Cycling MET: 7.5
- Strength Training MET: 5.0 | Swimming: 8.0 | Yoga: 3.0
- Intensity Multiplier: Adjusts ±25% for high/low intensity.
- Benefit: Accurate calorie expenditure without requiring smartwatches.

Biometric Health Metrics

- BMI (Body Mass Index): Weight (kg) / [Height (m)]² with clinical classifications (Underweight, Normal, Overweight, Obese).
- BMR (Mifflin-St Jeor): Calculates exact basal energy needed at rest based on age, gender, height, and weight.
- TDEE: Calculates total daily energy expenditure based on user workout frequency.
- Nutritional Calorie Targets: Recommends exact daily calorie deficit/surplus for fat loss or bulking.

DEMONSTRATION

Application Workflow & User Experience

Step-by-Step User Journey

- 1. Authentication: Fast signup or 1-click Demo Account login.
- 2. Dashboard Hub: Instant visibility into 7-day workouts, energy burned, active time, and current BMI badge.
- 3. Log Workout: User selects activity, types duration, clicks 'Auto' for instant calorie estimation, and saves.
- 4. Real-Time Sync: Workouts and calories instantly update dashboard KPI widgets and goals.

Interactive Analytics & Export

- Progress Page: Shows body weight curve dropping toward target weight line.
- Fitness Goals: Visual progress bars auto-fill with every logged session.
- Data Export: 1-Click 'Export CSV' generates a full Excel-compatible backup of all workouts.
- Profile Customization: Update targets and view recalculation of BMR and TDEE.

CONCLUSION

Conclusion & Future Enhancements

Project Summary & Key Results

• Full Fulfillment: All project requirements (Workout logging, progress monitoring, goal setting) successfully realized.

• Pure Tech Stack: Built cleanly with PHP 8, MySQL 8, Bootstrap 5, and Chart.js.

• Robust Architecture: Secure authentication, relational integrity, and automated calculation models.

• User-Centric UI: High responsiveness with intuitive, modern UI design.

Future Scope & Enhancements

• Wearable IoT Integration: Sync with Google Fit, Apple Health, and Garmin APIs.

• AI Coaching: Personalized workout recommendations using machine learning.

• Social Leaderboard: Community workout challenges with friends.

• Native Mobile App: Wrapper using Flutter or React Native.

• Q&A Session: Thank you! Any questions?